Robert A. Cochran III

Technical engineering leader | security systems, Kubernetes, software verification, AI-agent infrastructure

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PROFILE

Technical engineering leader and security systems researcher with a PhD in computer science and a publication record across NDSS, NSDI, POPL, and TISSEC. Work spans symbolic verification of remote client behavior, program synthesis, Kubernetes runtime security, eBPF-based observability, enterprise product architecture, engineering management, customer/design-partner strategy, and current ACS agentic foundations: secure harness primitives, skills strategy, support triage automation, internal hackathon leadership, and CI-tested agent workflows.

INTERESTS

Runtime security and behavioral detection, Kubernetes / OpenShift / RHACS, eBPF observability, Software verification and formal methods, Symbolic execution, Program synthesis, AI-agent safety and secure harnesses, Policy-as-code and detection layering, Developer/security workflows

EXPERIENCE AND RESEARCH RECORD

IBM / Red Hat - Red Hat Advanced Cluster Security (RHACS / ACS) — Staff Engineer / Engineering Manager

San Francisco, CA / Remote | 2021 - Present

- Lead runtime-security product and architecture work across RHACS process activity, network security, file activity, behavioral baselines, runtime anomaly detection, Kubernetes-native policy workflows, and analyst/developer experience.
- Drive eBPF-based runtime visibility design for OpenShift/Kubernetes environments, including collection semantics, policy exceptions, performance constraints, platform support, and low false-positive detection goals.
- Lead ACS agentic foundations: internal hackathon, CI tooling, support triage bots, skills strategy, secure harness primitives, and developer paths from local sandbox to OpenShift execution.
- Build and guide secure AI-agent infrastructure patterns through harness-openshell: declarative agent workflows, provider/credential boundaries, inference routing, policy-aware sandbox execution, and repeatable CI validation.
- Work with large enterprise customers and design partners to translate deployment constraints into roadmap decisions around scale, compliance, architecture, performance, IBM Power/Z support, and field-support workflows.
- Managed distributed engineering teams while remaining close to architecture and implementation; promoted L4 and L5 engineers through scope design, calibrated feedback, and career planning.
- Shaped OpenShift-first RHACS strategy around console integration, RBAC alignment, GitOps/policy-as-code, developer experience, runtime observability, and AI-assisted workflows.
- Outputs from this period:
- Proof object: harness-openshell - declarative configuration harness for OpenShell AI agent sandboxes.
- Patent: US 11,811,804, "System and method for detecting process anomalies in a distributed computation system utilizing containers"; Red Hat assignee; named inventor.
- Public talk: eBPF 101: implementing security and monitoring in Kubernetes.

StackRox, Inc. (acquired by Red Hat) — Member of Technical Staff / Senior Software Engineer / Staff Software Engineer

Mountain View, CA | May 2017 - 2021

- Joined an early-stage Kubernetes security startup and built runtime detection and enforcement features for containerized workloads and Kubernetes clusters.

- Advanced core runtime-security capabilities including baseline-driven detection, process and network visibility, and connections between observed runtime behavior and security policy.
- Worked across product, engineering, and customer-driven security cases with large design partners, helping mature the platform through acquisition by Red Hat.
- Contributed through the StackRox/RHACS product arc from early startup product to enterprise OpenShift security portfolio; product context included $0 to $50M+ revenue and 100+ releases.
- Outputs from this period:
- Company milestone: StackRox acquisition by Red Hat.
- Product: Red Hat Advanced Cluster Security for Kubernetes.
- Public talk: Listen to your Engine: Unearthing Security Signals from the Modern Linux Kernel. BSidesSF 2018.

University of North Carolina at Chapel Hill — Research Assistant, Computer Security and Software Verification

Chapel Hill, NC | 2008 - 2016

- Developed symbolic client verification: server-side methods for checking whether observed client network traffic could have been produced by sanctioned software.
- Built and evaluated symbolic-execution systems for online games, cryptographic clients, and distributed applications, including OpenSSL/Heartbleed-style client-misbehavior detection.
- Investigated data-driven symbolic execution using instruction traces, clustering, and novel metrics; refactored KLEE symbolic execution internals for thread-level parallelism and many-core execution.
- Research outputs included NDSS 2010 Best Paper, NDSS 2013, NSDI 2017, ACM TISSEC 2011, and dissertation work on remote client behavior.
- Outputs from this period:
- Dissertation: Symbolic Verification of Remote Client Behavior in Distributed Systems.
- A System to Verify Network Behavior of Known Cryptographic Clients. NSDI 2017.
- Toward Online Verification of Client Behavior in Distributed Applications. NDSS 2013.
- Server-side Verification of Client Behavior in Online Games. NDSS 2010 Best Paper; ACM TISSEC 2011 journal version.

Microsoft Research - Research in Software Engineering (RiSE) — Research Intern with Dr. Ben Livshits

Redmond, WA | Summer 2013

- Designed a program-synthesis workflow combining symbolic automata, genetic programming, and online workers to improve regular expressions from examples and human feedback.
- Outputs from this period:
- Publication: Program Boosting: Program Synthesis via Crowd-Sourcing. POPL 2015.
- Patent: US 9,753,696, "Program boosting including using crowdsourcing for correctness"; Microsoft Technology Licensing assignee; named inventor.

Intel Corporation - Visual Computing Group — Software Engineer

Hillsboro, OR | 2006 - 2007

- Built graphics-hardware analysis tooling to interpret, record, and replay Direct3D/OpenGL API calls for performance analysis of prototype visual-computing hardware.

Clemson University — Undergraduate Researcher

Clemson, SC | 2005 - 2006

- Designed and implemented GPU-based rendering research on second-order illumination and built network-simulation tooling for DOCSIS MAC-layer analysis.
- Outputs from this period:

- Publication: Second-order Illumination in Real-Time. ACM Southeast 2007; Best Paper.

EDUCATION

- **PhD, Computer Science**, University of North Carolina at Chapel Hill, 2016.
- Thesis: Symbolic Verification of Remote Client Behavior in Distributed Systems
- Advisor: Prof. Michael K. Reiter
- **BS, Computer Science, Magna Cum Laude**, Clemson University, 2006.
- Outstanding Senior in Computer Science; Donald A. Norton Computer Science Scholarship; Upsilon Pi Epsilon Honor Society

PUBLICATIONS

- A. Chi, R. Cochran, M. Nesfield, M. K. Reiter, C. Sturton. ”A System to Verify Network Behavior of Known Cryptographic Clients.” USENIX NSDI, 2017. *Origin: UNC.*
- R. Cochran, L. D'Antoni, B. Livshits, M. Veanes. ”Program Boosting: Program Synthesis via Crowd-Sourcing.” POPL, 2015. *Origin: Microsoft Research.*
- R. Cochran, M. K. Reiter. ”Toward Online Verification of Client Behavior in Distributed Applications.” NDSS, 2013. *Origin: UNC.*
- D. Bethea, R. Cochran, M. K. Reiter. ”Server-side Verification of Client Behavior in Online Games.” ACM TISSEC, 2011. *Origin: UNC.*
- D. Bethea, R. Cochran, M. K. Reiter. ”Server-side Verification of Client Behavior in Online Games.” NDSS, 2010. Best Paper. *Origin: UNC.*
- R. Cochran, J. Steele. ”Second-order Illumination in Real-Time.” ACM Southeast, 2007. Best Paper. *Origin: Clemson.*

PATENTS

- US 11,811,804. ”System and method for detecting process anomalies in a distributed computation system utilizing containers.” Red Hat, Inc.; filed Dec. 15, 2020; issued Nov. 7, 2023. Named inventor. *Origin: StackRox / Red Hat.*
- US 9,753,696. ”Program boosting including using crowdsourcing for correctness.” Microsoft Technology Licensing LLC; filed Mar. 14, 2014; issued Sept. 5, 2017. Named inventor. *Origin: Microsoft Research.*

AWARDS

- Best Paper Award, Network and Distributed System Security Symposium (NDSS), 2010.
- Best Paper Award, ACM Southeast Regional Conference, 2007.
- Outstanding Senior in Computer Science, Clemson University, 2006.
- Donald A. Norton Computer Science Scholarship, Clemson University, 2006.
- Upsilon Pi Epsilon Honor Society, Clemson University, 2005.

SERVICE

- External reviewer, ACM Conference on Data and Application Security and Privacy, 2013-2014.
- External reviewer, Network and Distributed System Security Symposium, 2013-2014.

TALKS AND PRESENTATIONS

- Listen to your Engine: Unearthing Security Signals from the Modern Linux Kernel. BSidesSF, 2018. (AllBSides archive)
- eBPF 101: implementing security and monitoring in Kubernetes. StackRox / Red Hat community office hours.

- Toward Online Verification of Client Behavior in Distributed Applications. NDSS, San Diego, CA, February 2013.
- Server-side Verification of Client Behavior in Online Games. NDSS, San Diego, CA, February 2010.
- Introduction to General Purpose GPU (GPGPU) Programming. ACM Southeast Regional Conference, March 2007.

TECHNICAL EXPERTISE

- Languages: Go, C, C++, Python, Rust, Bash, Java, C#, R
- Platforms: Kubernetes, OpenShift, RHACS, Docker, Linux, IBM Power, IBM Z
- Security / systems: eBPF, runtime detection, behavioral baselines, policy-as-code, LLM gateways, MCP, secure sandboxes, support automation, agent workflows
- Research / methods: symbolic execution, software verification, formal methods, program synthesis, SMT/SAT solvers, LLVM, KLEE

Last updated: June 2026.